

T R U T H C H E C K

Fact-checking Health Claims on YouTube.

An LLM agent that distinguishes settled medical consensus from genuine scientific debate with citations.



Final presentation

Sayalee Damle

Why a fact-checker for health videos

THE PROBLEM

- Claims made by YouTube videos might be misleading
- Need to fact check to decide whether a video is legit or not
- Fact check should be done with authoritative sources like:



PubMed

peer-reviewed studies, via E-utilities API



Cochrane (through PubMed)

systematic reviews and meta-analyses



CDC / WHO / NIH

official guidance and consensus statements

APPROACH

An agent, not a classifier.

Given a YouTube URL:

- TruthCheck extracts every checkable health claim
- Retrieves real peer-reviewed evidence
- Classifies each claim against scientific consensus

THE DIFFERENTIATOR

Clear consensus gets a verdict.

Active scientific debate gets both sides.

Six videos, one preprocessing finding – included in midterm

The corpus is clean and on-topic — the only real signal of trouble is caption punctuation on auto-generated transcripts.

6 videos

Huberman Lab + Nutrition Made Simple. ~72k words, 6 hours of long-form health content.

100%

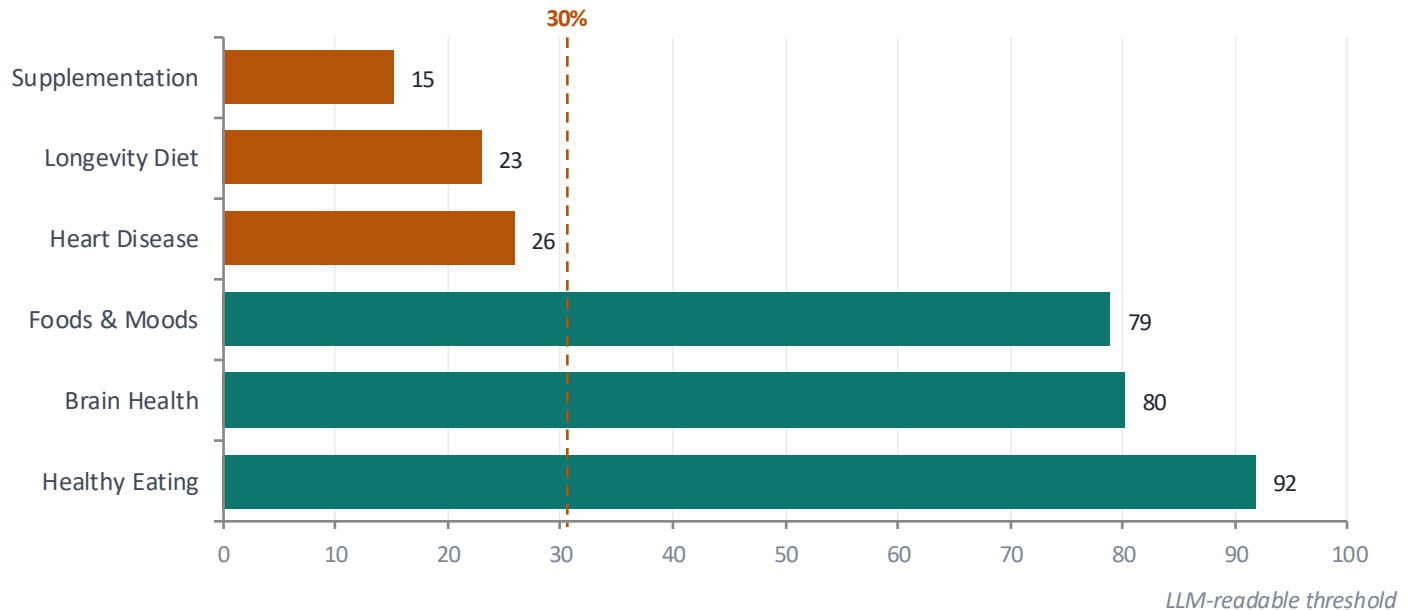
FIT 1M CONTEXT WINDOW
Largest video: ~29k tokens. No chunking needed for Gemini 2.5 Pro.

3 / 6

POORLY PUNCTUATED
Auto-generated captions drop to 15–26% punctuated segments — preprocessing step required.

PUNCTUATION RATE BY VIDEO

Manual captions vs. auto-generated

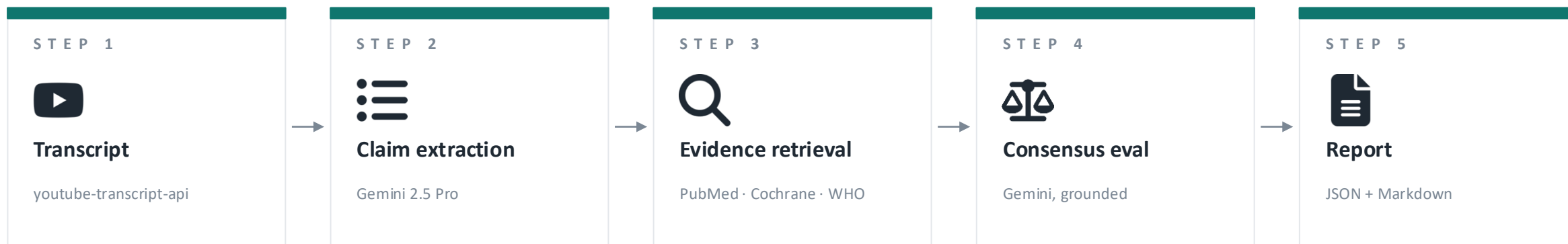


A 5-stage Gemini pipeline, grounded in real papers

Each stage is a focused LLM call or a fan-out to PubMed via NCBI E-utilities.

The stages don't talk to each other directly; they hand off objects.

Pydantic schemas form the contract between stages

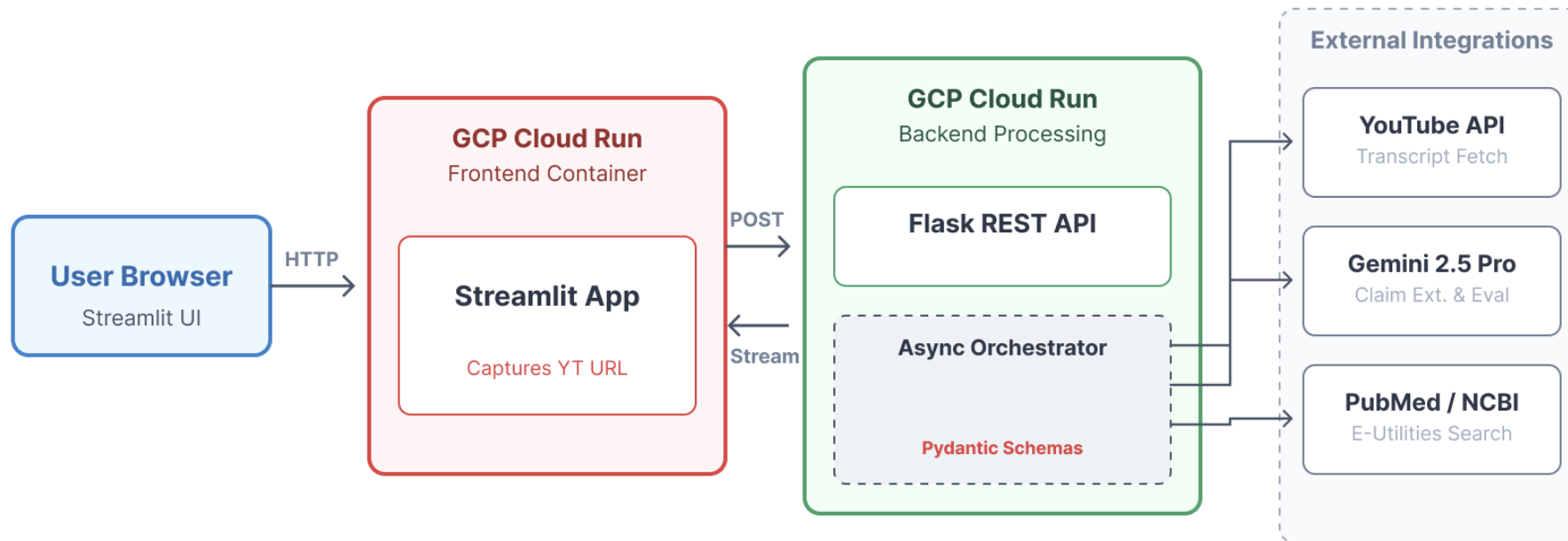


LIVE TEST — "WHAT, WHY & HOW OF HEALTHY EATING" · 2 CLAIMS · 42 SECONDS

Claim	Verdict	Why this is the right call
"Nutrition is a leading factor in disease risk."	Unverifiable	Retrieved GBD papers describe methodology, not rankings. Model correctly declined to overclaim.
"Lifestyle fundamentals prevent many medical problems."	Unverifiable	Retrieved WHO/NIH papers tangentially related. Model grounded, didn't hallucinate consensus.

The deployed system, end to end

TruthCheck · Final Deployed Solution Architecture



DEPLOYED ON Google Cloud Run · Docker images · scales to zero · no persistent database · GEMINI_API_KEY via env var

What the user sees

API URL
Configuration
Field

 **TruthCheck**
Health & Nutrition Fact-Checker for YouTube

API URL

https://truthcheck-api-nidvsh553a-uc.a.run.

Max claims to evaluate

10

Verdict key

✓	Supported by consensus
✗	Contradicted by consensus
⚠	Contested / actively debated
?	Unverifiable
□	Out of scope

Max Claims
Slider
Guardrail

Contextual Help

 **TruthCheck**

AI-powered health & nutrition fact-checker for YouTube videos

Paste a YouTube URL below. TruthCheck extracts every health claim from the transcript and evaluates each one against peer-reviewed scientific literature.

https://www.youtube.com/watch?v=lb9JmKG7Q9c

Analyze ▶

▼  Paste transcript manually (use this if URL fetch fails)

welcome everyone we've got dr gil carvalho from nutrition made simple his channel is about to cross over 100 000 subscribers and i think it's awesome to have him on the show today so welcome gil thanks for having me so today we're hoping to go through diet supplements medications fasting a whole bunch of things to hopefully extend health span and lifespan so i think we'll start off with diet in particular looking at popular diet myths so gil what sort of things have you come across online that is quite prevalent that doesn't necessarily have that much evidence behind it every day is a new one it's uh there's a by this point there's nothing i haven't seen uh the only thing i joke that the only thing i haven't seen is the all junk food diet uh being promoted as healthy although that's already what most americans are doing anyway so uh but we haven't seen that that uh fitness trend uh what are some only yeah i mean there's there's everything under the sun um we see a lot of uh we see a lot of so that on social media i think things are separated into diet camps and so people kind of find a diet that they like or that makes them

Analysis complete — 10 claims evaluated in 130s

Video: <https://www.youtube.com/watch?v=lb9JmKG7Q9c>

YouTube URL
Input Field

Action Button

If API Fails,
paste the
transcript
here (can be
done
locally)

Walk Through

<https://truthcheck-app-nidvsh553a-uc.a.run.app>

TruthCheck
Health & Nutrition Fact Checker for YouTube

API URL
<https://truthcheck-api-nidvsh553a-uc.a.run.app>

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Video: <https://www.youtube.com/watch?v=ib9JmKG7Q9c>

If the API Fails

<https://www.youtube.com/watch?v=ib9JmKG7Q9c> **Analyze**

☐ Fetching transcript and evaluating claims — please wait...

☒ Paste transcript manually (use this if URL fetch fails)

think that should be encouraged um but as you say we can't just throw out the baby with the bath water we need to make sure that we're looking at the whole person and if a particular diet is making you know ldl cholesterol april b go up if it's you know causing issues with blood sugar levels i think we need to look at the whole picture and just you know take a step back and and i found it online i found it a little bit tricky to talk to some people about that but i think you do a fantastic job on your um on your channel about you know presenting the entire um the entire picture so that people can make up their own mind and i think at the end of the day that's all we can do is just present people with you know the latest information and it's up to them to make their own their own decision so i'm curious what um what you see about that what you think about that strategy that's a perfect

☐ Fetching transcript and evaluating claims — please wait...

1. ? Unverifiable --71s — People who consume large amounts of red meat or processed red meat have a higher risk of d...

Claim: People who consume large amounts of red meat or processed red meat have a higher risk of developing cancer.

Context: if the question is do people who eat a lot of red meat or do people who eat a lot of processed red meat have a higher risk of getting cancer i think the answer to that question is yes but it's probably less because of the meat...

Explanation: The provided evidence is insufficient to evaluate the claim. One source discusses the pathophysiology of uterine fibroids, and the other examines a specific diet's effect on breast cancer recurrence. Neither of these sources contains information about the relationship between red or processed meat consumption and the general risk of developing cancer.

Citations:

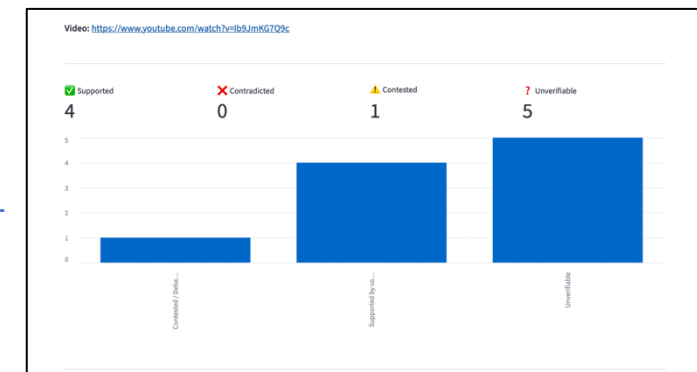
- [PubMed PMID:27335259](#) — Uterine fibroids (also known as leiomyomas or myomas) are common clonal neoplasms of the uterus. Fibroids have both smooth muscle and fibroblast components, in addition to a substantial amount of fibrous extracellular matrix, which all contribute to the pathogenetic process.
- [PubMed PMID:37847493](#) — The DIANA-5 randomized controlled trial assessed the effectiveness of a diet based on Mediterranean and macrobiotic traditions (macro-Mediterranean diet) in reducing breast cancer recurrence.

Claim-by-claim breakdown (10)

- 1. Supported by consensus** --122s — Elevated cholesterol and triglycerides are risk factors for disease even for individuals o...
- 2. Contested / Debated** --130s — ApoB and cholesterol levels are important risk factors for disease, even for individuals o...
- 3. ? Unverifiable** --442s — The Mediterranean diet has the most evidence behind it from large, long-term clinical tria...
- 4. Supported by consensus** --515s — Diets rich in fruits, vegetables, unprocessed plants, legumes, whole grains, nuts, seeds, ...
- 5. Supported by consensus** --540s — Excessive added sodium is consistently associated with high blood pressure and a higher ri...
- 6. Supported by consensus** --550s — Consuming unsaturated fats is associated with better health outcomes compared to consuming...
- 7. ? Unverifiable** --663s — The body of scientific evidence indicates that whole grains are beneficial for health.
- 8. ? Unverifiable** --720s — Individuals with type 2 diabetes may experience significant spikes in blood glucose after ...
- 9. ? Unverifiable** --900s — The Recommended Dietary Allowance (RDA) for protein is 0.8 grams per kilogram of body weig...
- 10. ? Unverifiable** --928s — Consuming more protein than the RDA is beneficial for retaining muscle mass in individuals...

Download JSON report

Download Markdown report



Three real problems I faced



YouTube API Doesn't work on Cloud

PROBLEM

YouTube blocks any requests made real-time from a cloud, which is creating an hinderance in the project.

FIX

Add Fault Tolerance where the user can add transcripts manually onto to UI.



Automatic Evaluation is a problem

PROBLEM

Evaluating multi-stage pipelines like TruthCheck is notoriously difficult because standard traditional software tests

FIX

Use LLM as a Judge or create a Golden Dataset



Tangential evidence retrieval

PROBLEM

Verdicts skewed `unverifiable`.

FIX

Honest framing: the model correctly refuses to overclaim.

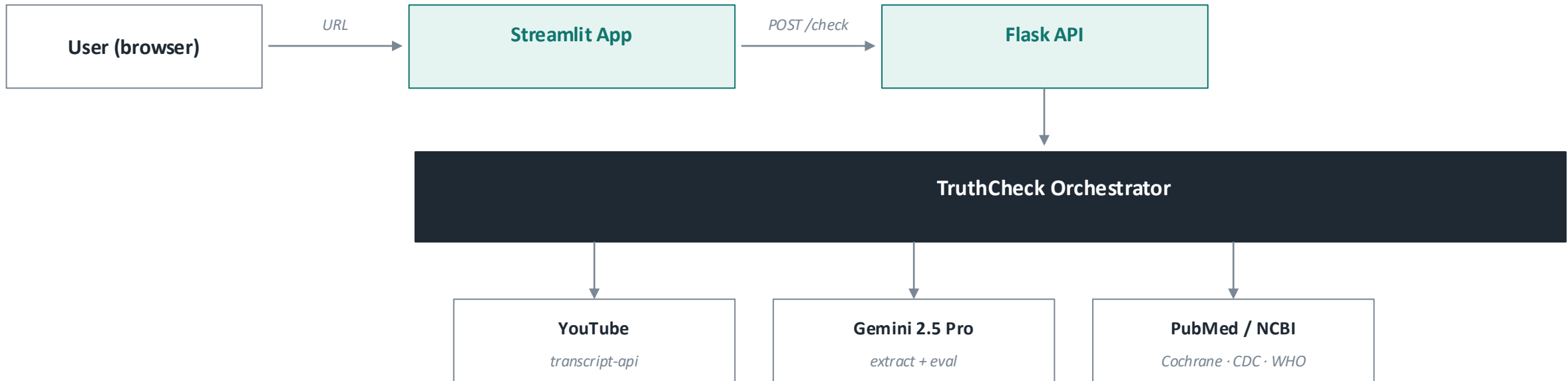
T R U T H C H E C K

THANK YOU!

GitHub repo: <https://github.com/sayaleedamle/418-final-project>

The deployed system, end to end

Two containerized services on Cloud Run: a Streamlit UI and a Flask REST API. Stateless, scales to zero, no database. The orchestrator calls Gemini twice (extract, evaluate) and PubMed once per claim.



DEPLOYED ON Google Cloud Run · Docker images · scales to zero · no persistent database · GEMINI_API_KEY via env var

Walk Through



Analyze ▶

○ Fetching transcript and evaluating claims — please wait...



TruthCheck

Health & Nutrition Fact-Checker for YouTube

API URL

http://localhost:8080

Max claims to evaluate

5

Verdict key

✔ Supported by consensus

✘ Contradicted by consensus

⚠ Contested / actively debated

❓ Unverifiable

☐ Out of scope

Evidence sourced from PubMed, Cochrane Reviews, FOD, WUOL & NIH

TruthCheck

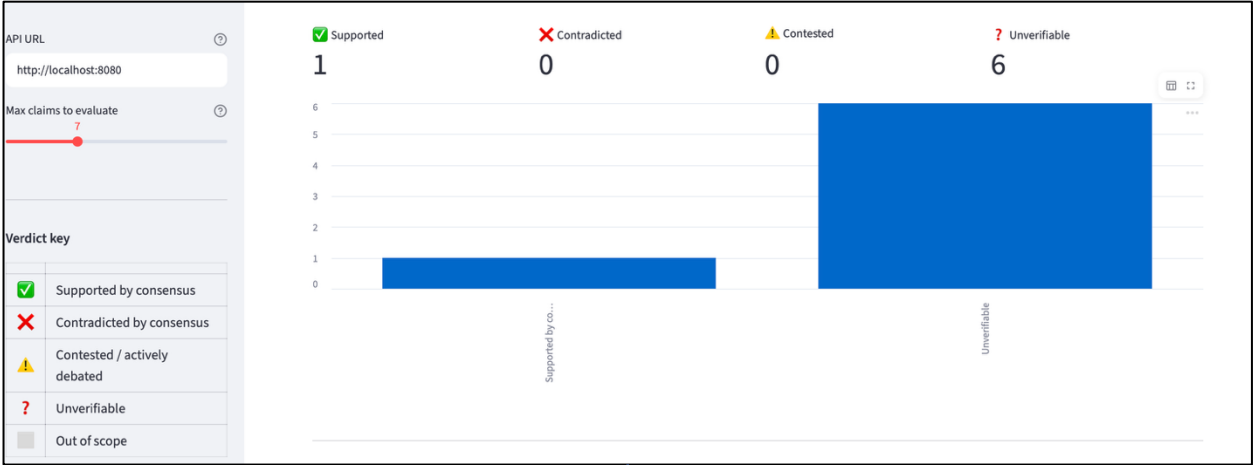
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https://www.youtube.com/watch?v=...

Analyze ▶

Enter a health or nutrition YouTube video URL and click Analyze ▶. Processing takes 1-3 minutes depending on transcript length.



Claim-by-claim breakdown (7)

> 1. ❓ Unverifiable · ~301s — The vagus nerve is the 10th cranial nerve.

> 2. ❓ Unverifiable · ~313s — The vagus nerve has branches that connect to the brain, stomach, intestines, heart, lungs,...

> 3. ❓ Unverifiable · ~357s — Neurons in the stomach can sense the presence of sugary foods independently of taste and s...

> 4. ❓ Unverifiable · ~368s — Sugar-sensing neurons in the gut send information up the vagus nerve to the brain, trigger...

> 5. ❓ Unverifiable · ~387s — Humans who cannot taste sugary food still develop a craving for it due to sugar sensors in...

> 6. ❓ Unverifiable · ~480s — Humans tend to eat until the brain perceives an adequate intake of amino acids, rather tha...

> 7. ✔ Supported by consensus · ~495s — Amino acids are the precursors from which neurochemicals in the brain are made.



1. ❓ Unverifiable · ~71s — People who consume large amounts of red meat or processed red meat have a higher risk of d...

Claim: People who consume large amounts of red meat or processed red meat have a higher risk of developing cancer.

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What the user sees

USER FLOW

- 1 Paste YouTube URL**
Plus a max-claims slider (1–20) to cap pipeline cost.
- 2 Watch progress live**
Spinner with stage-by-stage messaging while Gemini runs.
- 3 Read the summary**
4-metric row: supported · contradicted · contested · unverifiable.
- 4 Expand any claim**
Timestamp, verdict, explanation, real citation URLs.
- 5 Download JSON / MD**
Structured report for sharing or further analysis.

